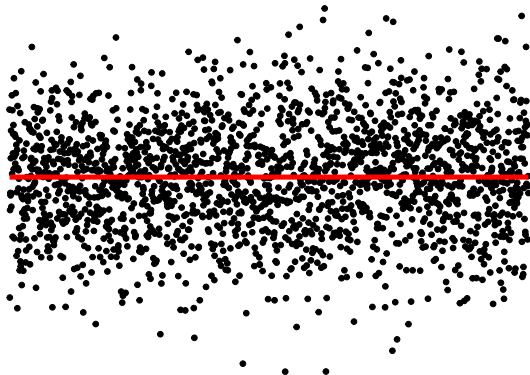
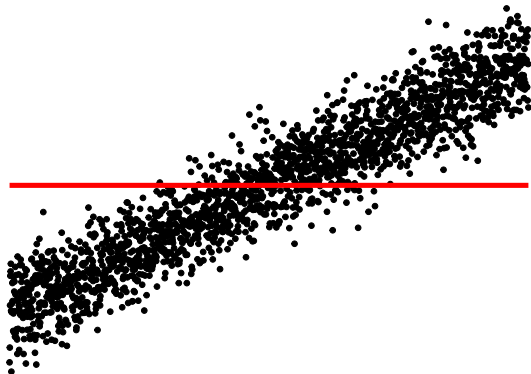


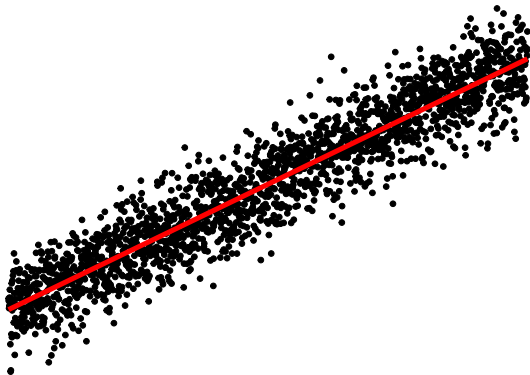
Tackling Global Issues: Change

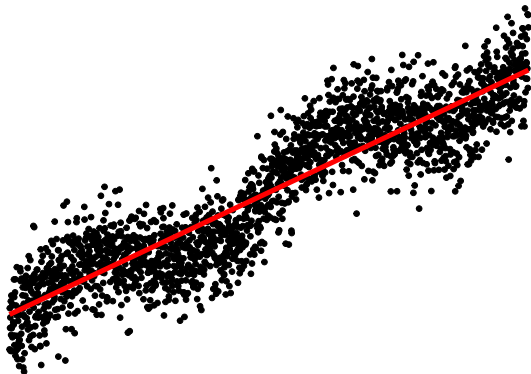
Rebecca Killick(r.killick@lancs.ac.uk)
Manchester Nov 2021

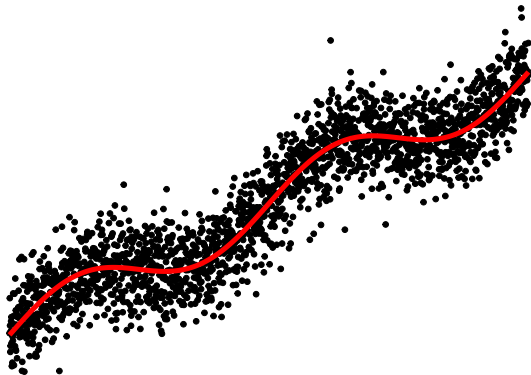


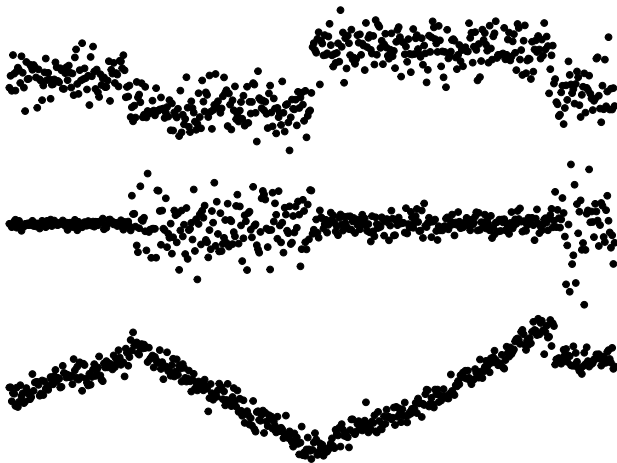










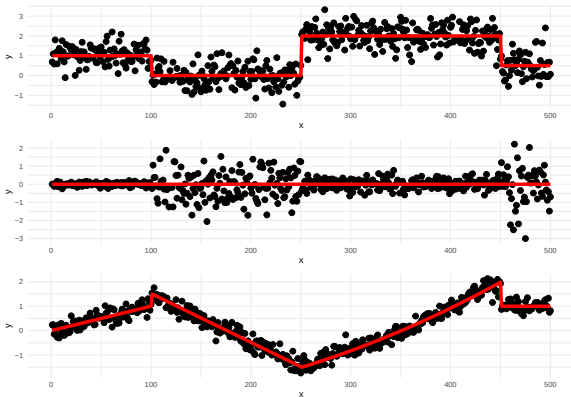


- What are changepoints?
- Tour of changepoint problems and applications
 - Speed & accuracy
 - Autocorrelation
 - Model choice
 - Multivariate
- Open problems
 - Autocorrelation
 - Multivariate
 - Sensitivity to data
 - Online inference

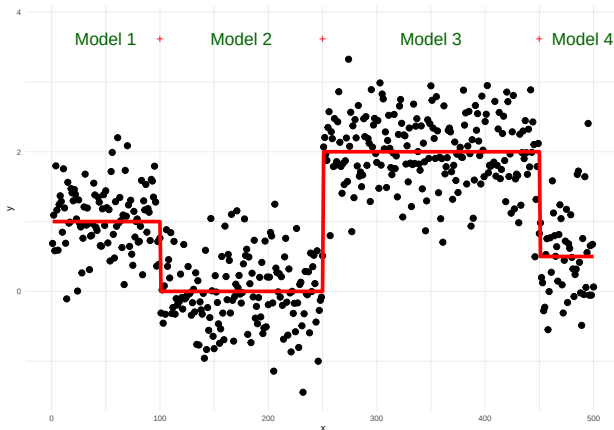
What are changepoints?

For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.



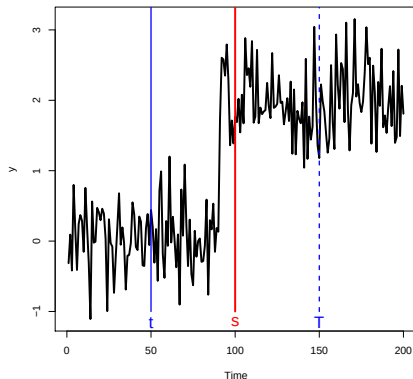
Problem



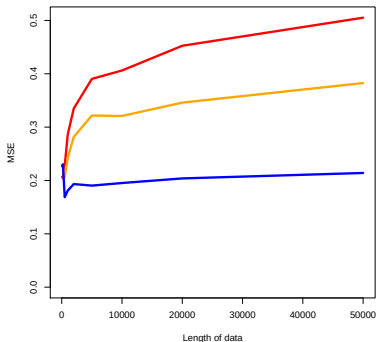
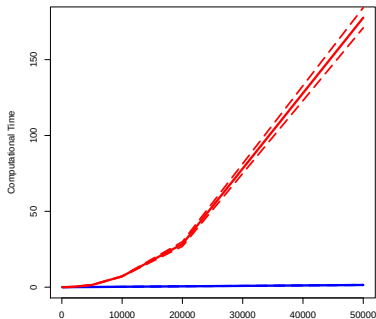
- How many changes?
- Where are the changes? 2^{n-1} possible solutions!



Computational complexity in 2010: $\mathcal{O}(n^2)$ exact, $\mathcal{O}(n \log n)$ approx.
 $\mathcal{O}(n^2)$ solution book keeps the last changepoint location. BUT ...



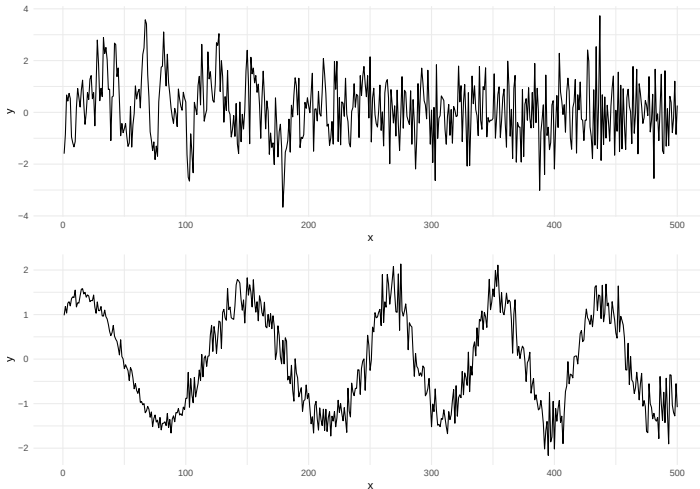
... resulting in an $\mathcal{O}(n)$ algorithm if changepoints occur regularly.



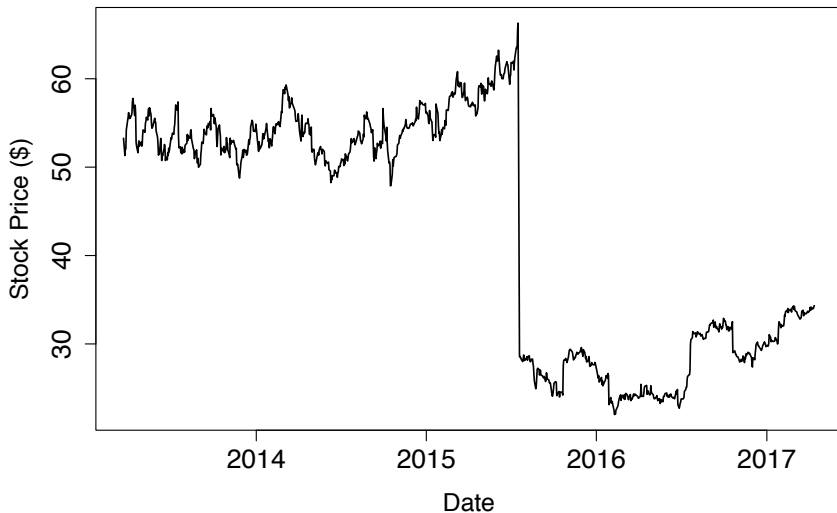
SPEED + ACCURACY = PELT!

More complicated changes

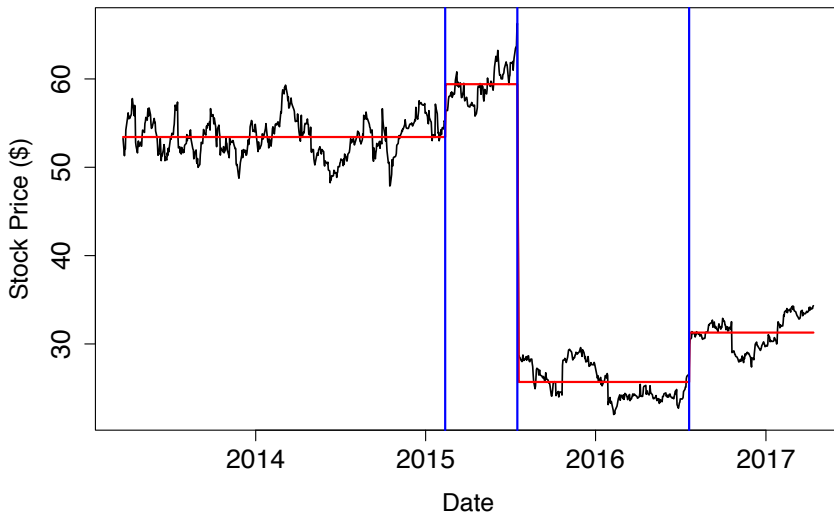
Mathematics
& Statistics



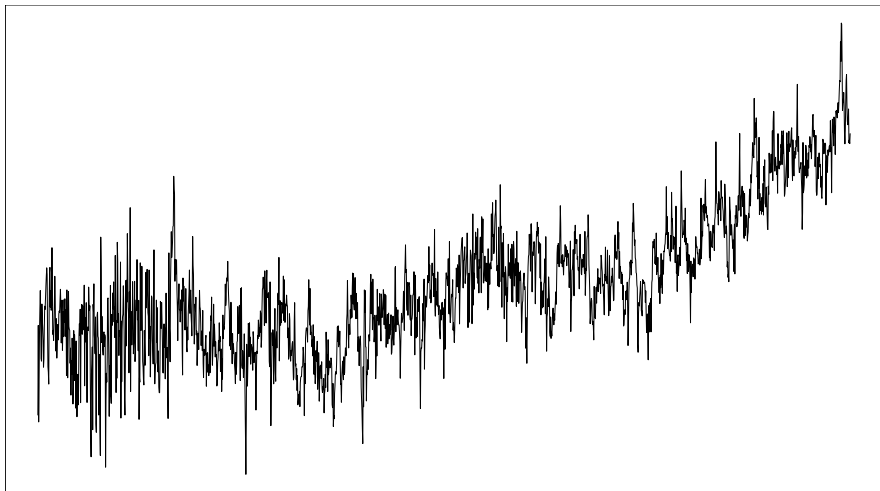
Changes are hard to identify with background correlation



Change in mean with non-stationary errors

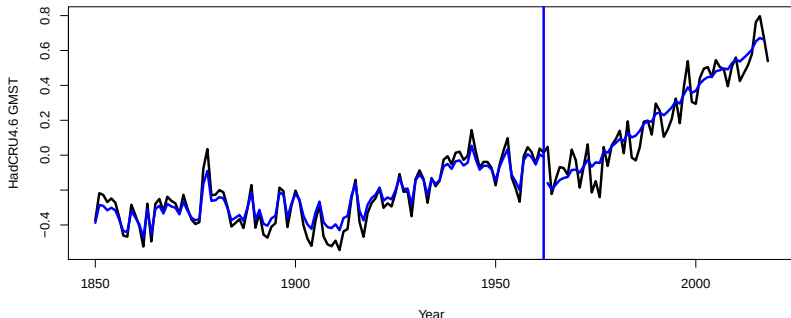


Natural correlation

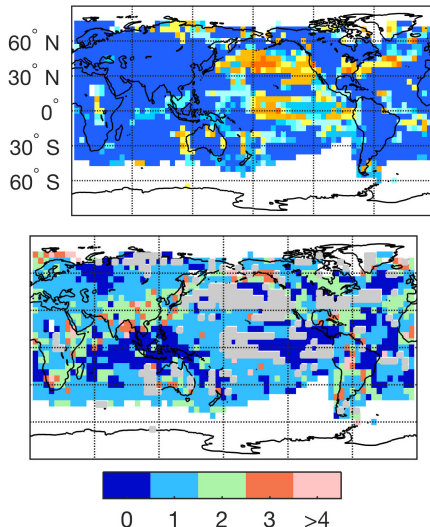




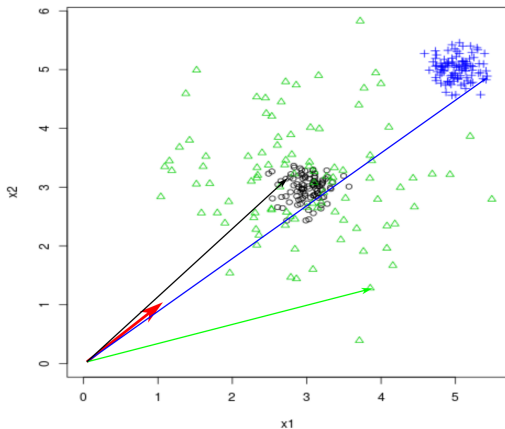
- Has there been a recent hiatus in global warming?
- Analysed several surface temperature time series estimates.



- Developed automatic model selection including trend, autocorrelation and changepoint detection
- This indicates no hiatus in all series.

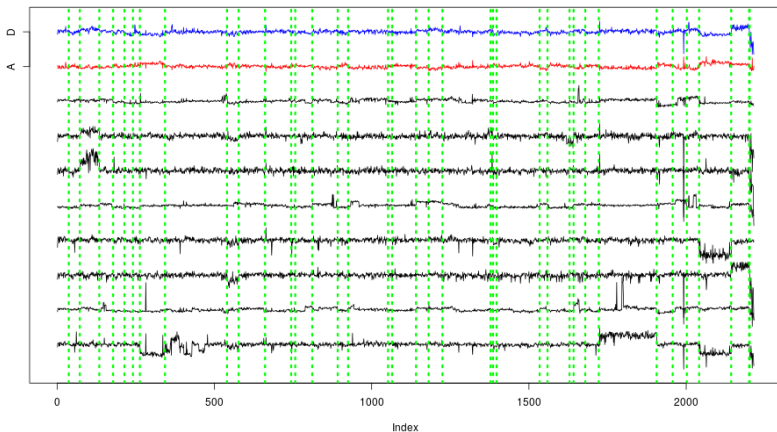


- Which genomic markers could be indications for disease?
- Mean and variance changes



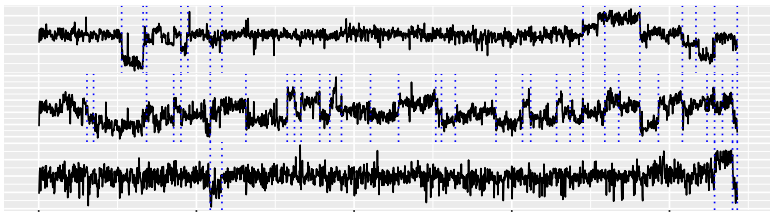


- Which genomic markers could be indications for disease?
- Use two transformations instead of one



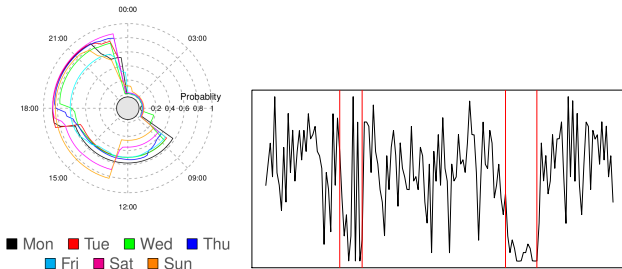


- Which genomic markers could be indications for disease?
- Only markers present in lots of patients are worth investigating.



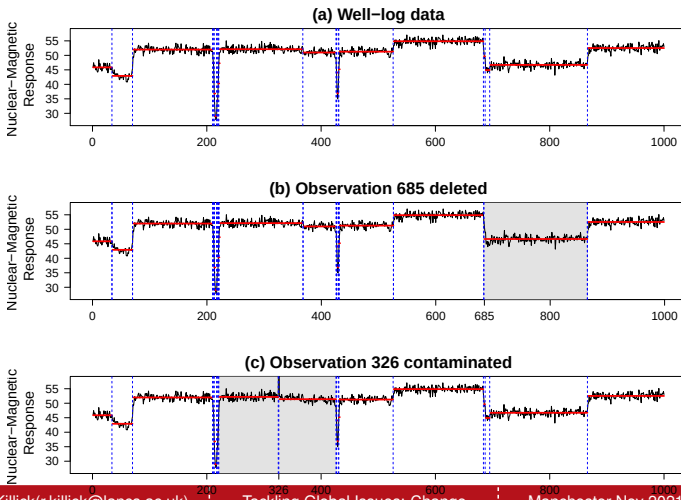
- Developed computationally efficient subset multivariate changepoint methodology.
- Only 10 changes are present in more than 60% of the patients.
- Over 50 detected by competitors.

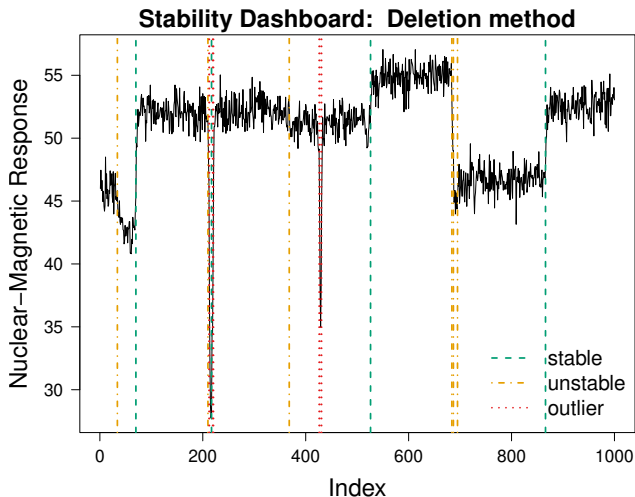
- Passively monitor patients at home
- Desire to have early intervention before crisis point is reached

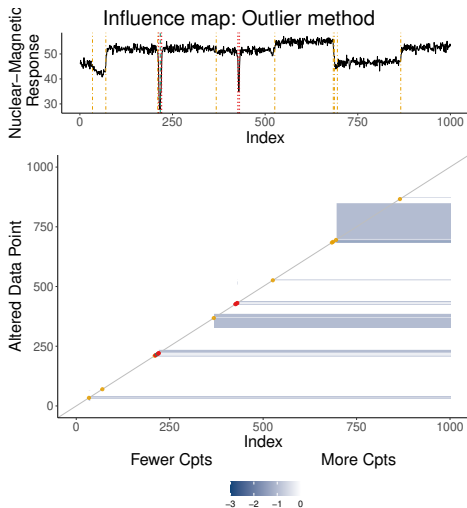


- View data as circular, e.g., each day.
- Developed circular changepoint detection to detect changes in activity levels within a day
- Changes in the detected routine signal early intervention

- Changepoint locations are sensitive to the observed data
- Some changepoint algorithms are more sensitive than others.







- Autocorrelation
- Multivariate data
- Model complexity
- Online changes

- Changepoints occur in most data
- Detecting and documenting them improves analyses
- Multivariate, online, big data mostly isn't a problem
- But there are still plenty of open problems in the field
- I enjoy working with messy real life data ...
- ... as often it sparks my next research challenge.

Preprints of all available at: www.lancs.ac.uk/~killick/pub.html

PELT: <https://doi.org/10.1080/01621459.2012.737745>

LMvsCpts: <https://doi.org/10.1007/s11222-017-9731-0> &
<https://doi.org/10.1002/env.2568>

Model Choice: <https://doi.org/10.1175/JCLI-D-17-0863.1> &
<https://doi.org/10.1002/qre.2712>

Multivariate: <https://doi.org/10.1038/s41598-019-46836-y> &
<https://doi.org/10.1007/s11222-020-09940-y>

Autocorrelation: <https://doi.org/10.1002/sta4.351>

Circular time: <http://doi.org/10.1111/rssc.12472>

Sensitivity (accepted in JCGS): <https://arxiv.org/abs/2107.10572>

Multivariate: <https://arxiv.org/abs/2108.07340>

Autocorrelation: <https://arxiv.org/abs/2102.10669>

Online: Email for preprint, code:

<https://github.com/grundy95/changepoint.forecast>

Change in variance with nonstationary errors:

<https://doi.org/10.1002/env.2576>

Changes in second order structure within nonstationary series:

<http://dx.doi.org/10.1214/13-EJS799>

Uncertainty of autocovariance changes:

<https://doi.org/10.1080/00401706.2014.902776>